A Graph Features

The features of the instances utilized in this study are detailed as follows:

- LOG_NUM_EDGES: The logarithm of the number of edges in the graph.
- DENSITY: The proportion of actual edges to the maximum possible number of edges (i.e., of a complete graph).
- LOGRATIO_EDGETONODES: The natural logarithm of the edge-to-node ratio.
- LOG_CHROMATIC_NUM: The logarithm of the minimum number of colours required to colour the graph such that no two adjacent nodes share the same colour.
- NORM_MIS: The size of the largest independent node-set, adjusted relative to the total number of nodes, showcasing the proportion of non-adjacent nodes.
- GRAPH_ASSORTATIVITY: Gauges node connection similarity based on node degree, with values near 1 suggesting similar degree connections.
- SPECTRAL_GAP: The difference between the largest and second-largest eigenvalues of the graph's normalized Laplacian matrix.
- LOG_LARGESTEIGVAL: The logarithm of the highest eigenvalue from the normalized Laplacian matrix.
- LOG_SECONDLARGESTEIGVAL: The logarithm of the normalized Laplacian matrix's second-largest eigenvalue.
- LOG_SMALLESTEIGVAL: The logarithm of the smallest nontrivial eigenvalue from the normalized Laplacian matrix.
- GIRTH: The length of the shortest loop in the graph.
- TRANSITIVITY: The likelihood ratio of nodes to form tightly connected clusters.

B GW-Related Features

The features related to the GW algorithm are introduced as follows:

- PERCENT_CUT: The ratio of the optimal cut value derived from the semidefinite programming (SDP) solution to the overall number of edges in the graph.
- PERCENT_POSITIVE_LOWER_TRIANGULAR: Quantifies the share of positive entries in the lower triangular section of the factorized matrix.
- PERCENT_CLOSE1_LOWER_TRIANGULAR: The proportion of elements in the lower triangular of the factorized matrix that have absolute values under 0.1.
- PERCENT_CLOSE3_LOWER_TRIANGULAR: The percentage of elements in the lower triangular of the factorized matrix with absolute values smaller than 0.001.
- EXPECTED_COSTGW_OVER_SDP_COST: The proportion of the GW approach's average cost (over 1,000 random trials) relative to the maximum theoretical cost as determined by the SDP method.
- STD_COSTGW_OVER_SDP_COST: Quantifies the standard deviation of the outcomes from 1,000 random implementations of GW, normalized by the best possible cost determined through the SDP method.

C Machine Learning Classifiers

The models with hyperparameters constructed in the section of feature analysis with the machine learning pipeline are highlighted in the following:

- The classifier developed using `TPOT` for instances involving 20 nodes uses an ExtraTreesClassifier, which comprises a collection of 100 decision trees. This ensemble employs the entropy criterion to determine the quality of splits within the trees. Each tree considers approximately 35% of the features for node splitting. It also ensures that each leaf node has at least one sample, while at least nine samples are necessary to initiate further node splitting. The random_state parameter is set to 42, ensuring reproducibility.
- The classifier built with `TPOT` for 100-node instances is implemented as an ExtraTreesClassifier that features bootstrapping. The entropy criterion is used to evaluate split quality across the trees. It also requires 80% of the features to be considered for each tree split and a minimum of 19 samples for each leaf node. Additionally, a node split requires at least 5 samples to proceed. The ensemble includes 100 trees, and the random_state parameter is fixed at 42.

The models with hyperparameters constructed in the section of RQAOA as a high-performing heuristic with the machine learning pipeline are highlighted in the following:

- The classifier configured for 20-node instances using the `TPOT` library employs a sequential machine-learning pipeline. It begins with a StackingEstimator that deploys an MLPClassifier, characterized by a learning rate of 1.0 and a low regularization factor. This is complemented by another StackingEstimator, which utilizes an ExtraTreesClassifier. This classifier uses the entropy criterion method, avoids bootstrapping, and considers 65% of the features during each split, all while adhering to predefined parameters for sample splits and minimum leaf size. The data is then normalized using a StandardScaler to ensure the features are optimally scaled. Finally, the process culminates with a GaussianNB classifier.
- The classifier configured using `TPOT` for 100-node instances also employs a sequential machine-learning pipeline. Initially, a StackingEstimator incorporating a GaussianNB classifier makes preliminary predictions that feed into later stages. Following that, a SelectFwe feature selector uses the f_classif with an alpha of 0.012 to selectively maintain features. Subsequently, another StackingEstimator featuring a Decision Tree Classifier is used, which is tailored with the Gini index for node splitting, a cap of 7 on tree depth, a minimum of 9 samples per leaf, and at least 12 samples required for further splitting. A RobustScaler is then employed to adjust feature scales. The process concludes with a GaussianNB classifier.

D Important Features with Machine Learning Pipeline

Important score	Feature
0.1471 ± 0.0059	EXPECTED_COSTGW_OVER_SDP_COST
0.1370 ± 0.0084	SPECTRAL_GAP
0.0769 ± 0.0057	PERCENT_CLOSE1_LOWER_TRIANGULAR
0.0684 ± 0.0031	PERCENT_POSITIVE_LOWER_TRIANGULAR
0.0665 ± 0.0073	LOG_SECONDLARGESTEIGVAL
0.0528 ± 0.0042	LOG_LARGESTEIGVAL
0.0527 ± 0.0042	DENSITY
0.0521 ± 0.0056	LOG_NUM_EDGES
0.0508 ± 0.0051	PERCENT_CLOSE3_LOWER_TRIANGULAR
0.0434 ± 0.0068	LOGRATIO_EDGETONODES
0.0431 ± 0.0040	NORM_MIS
0.0413 ± 0.0036	GRAPH_ASSORTATIVITY
0.0408 ± 0.0042	TRANSITIVITY
0.0378 ± 0.0027	STD_COSTGW_OVER_SDP_COST
0.0369 ± 0.0049	PERCENT_CUT
0.0310 ± 0.0052	LOG_SMALLESTEIGVAL
0.0215 ± 0.0041	LOG_CHROMATIC_NUM
0.0000 ± 0.0000	GIRTH

Table A1: Important features with the classifier built on 20-node maximum cut instances.

Important score	Feature
0.3749 ± 0.0104	EXPECTED_COSTGW_OVER_SDP_COST
0.1588 ± 0.0196	LOG_SMALLESTEIGVAL
0.1281 ± 0.0052	PERCENT_CLOSE1_LOWER_TRIANGULAR
0.0648 ± 0.0081	PERCENT_CUT
0.0618 ± 0.0049	PERCENT_CLOSE3_LOWER_TRIANGULAR
0.0454 ± 0.0071	DENSITY
0.0322 ± 0.0054	GRAPH_ASSORTATIVITY
0.0239 ± 0.0049	LOG_LARGESTEIGVAL
0.0199 ± 0.0085	LOG_NUM_EDGES
0.0165 ± 0.0037	NORM_MIS
0.0153 ± 0.0036	TRANSITIVITY
0.0143 ± 0.0026	LOG_SECONDLARGESTEIGVAL
0.0126 ± 0.0061	LOGRATIO_EDGETONODES
0.0116 ± 0.0026	LOG_CHROMATIC_NUM
0.0096 ± 0.0030	SPECTRAL_GAP
0.0061 ± 0.0032	PERCENT_POSITIVE_LOWER_TRIANGULAR
0.0043 ± 0.0025	STD_COSTGW_OVER_SDP_COST
0.0000 ± 0.0000	GIRTH

Table A2: Important features with the classifier built on 100-node maximum cut instances.

E Important Features with RQAOA as a High-performing Heuristic

Important score	Feature
0.0665 ± 0.0641	STD_COSTGW_OVER_SDP_COST
0.0361 ± 0.0382	DENSITY
0.0355 ± 0.0389	NORM_MIS
0.0352 ± 0.0383	GRAPH_ASSORTATIVITY
0.0347 ± 0.0372	LOG_LARGESTEIGVAL
0.0345 ± 0.0376	LOG_CHROMATIC_NUM
0.0331 ± 0.0344	TRANSITIVITY
0.0319 ± 0.0201	SPECTRAL_GAP
0.0180 ± 0.0191	LOG_SECONDLARGESTEIGVAL
0.0082 ± 0.0051	LOG_SMALLESTEIGVAL
0.0031 ± 0.0034	PERCENT_POSITIVE_LOWER_TRIANGULAR
0.0031 ± 0.0039	LOG_NUM_EDGES
0.0028 ± 0.0049	PERCENT_CUT
0.0012 ± 0.0025	LOGRATIO_EDGETONODES
0.0010 ± 0.0048	PERCENT_CLOSE1_LOWER_TRIANGULAR
0.0002 ± 0.0051	PERCENT_CLOSE3_LOWER_TRIANGULAR
0.0000 ± 0.0000	GIRTH
-0.0002 ± 0.0029	EXPECTED_COSTGW_OVER_SDP_COST

Table A3: Important features with the classifier built on 20-node maximum cut instances demonstrate RQAOA's high performance, underscored by its significant permutation importance.

Important score	Feature
0.0243 ± 0.0379	EXPECTED_COSTGW_OVER_SDP_COST
0.0166 ± 0.0339	DENSITY
0.0114 ± 0.0057	LOGRATIO_EDGETONODES
0.0066 ± 0.0110	NORM_MIS
0.0036 ± 0.0106	LOG_SECONDLARGESTEIGVAL
0.0028 ± 0.0047	LOG_NUM_EDGES
0.0023 ± 0.0048	LOG_LARGESTEIGVAL
0.0011 ± 0.0020	PERCENT_POSITIVE_LOWER_TRIANGULAR
0.0010 ± 0.0052	LOG_CHROMATIC_NUM
0.0003 ± 0.0027	STD_COSTGW_OVER_SDP_COST
0.0003 ± 0.0004	SPECTRAL_GAP
0.0001 ± 0.0002	TRANSITIVITY
0.0000 ± 0.0000	GIRTH
-0.0001 ± 0.0004	PERCENT_CLOSE1_LOWER_TRIANGULAR
-0.0001 ± 0.0003	PERCENT_CLOSE3_LOWER_TRIANGULAR
-0.0002 ± 0.0004	PERCENT_CUT
-0.0002 ± 0.0008	GRAPH_ASSORTATIVITY
-0.0153 ± 0.0118	LOG_SMALLESTEIGVAL

Table A4: Important features with the classifier built on 100-node maximum cut instances demonstrate RQAOA's high performance, underscored by its significant permutation importance.